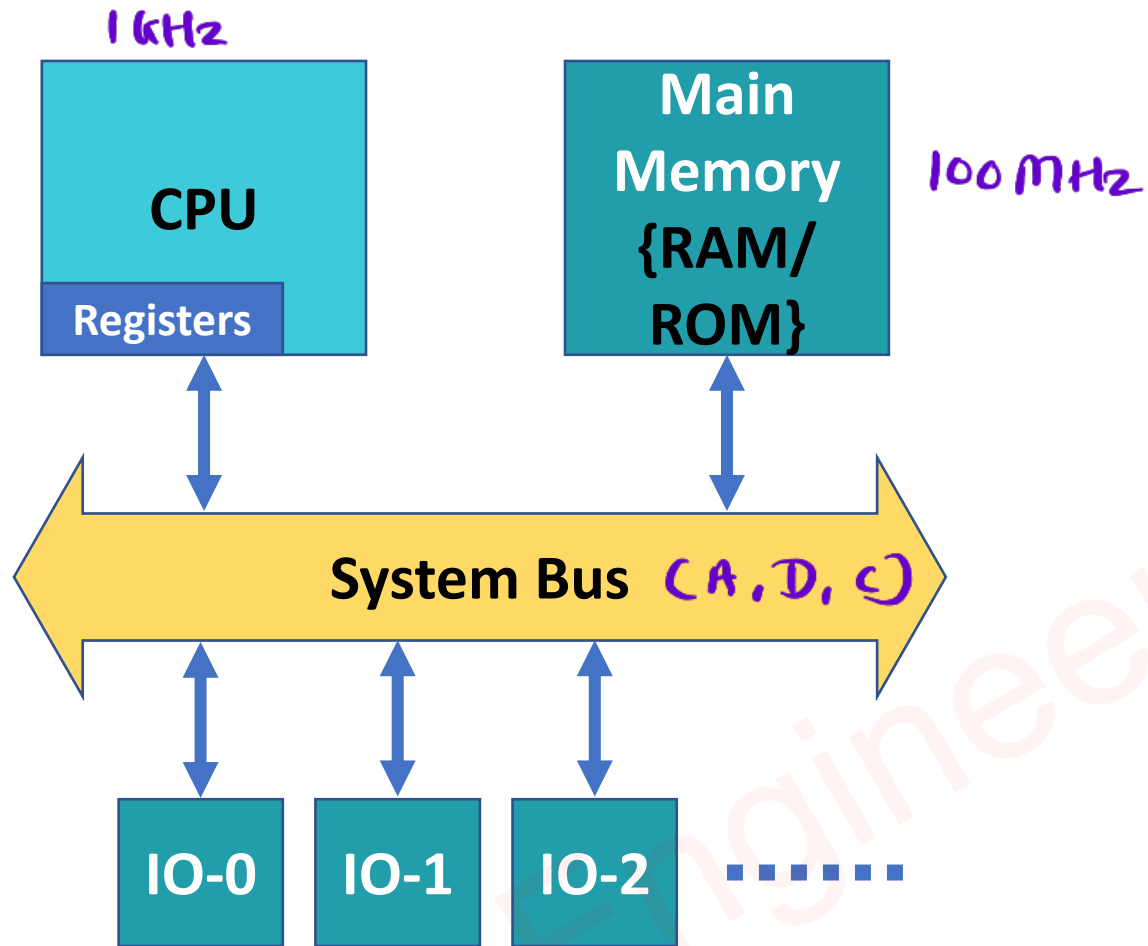


Input Output Organization

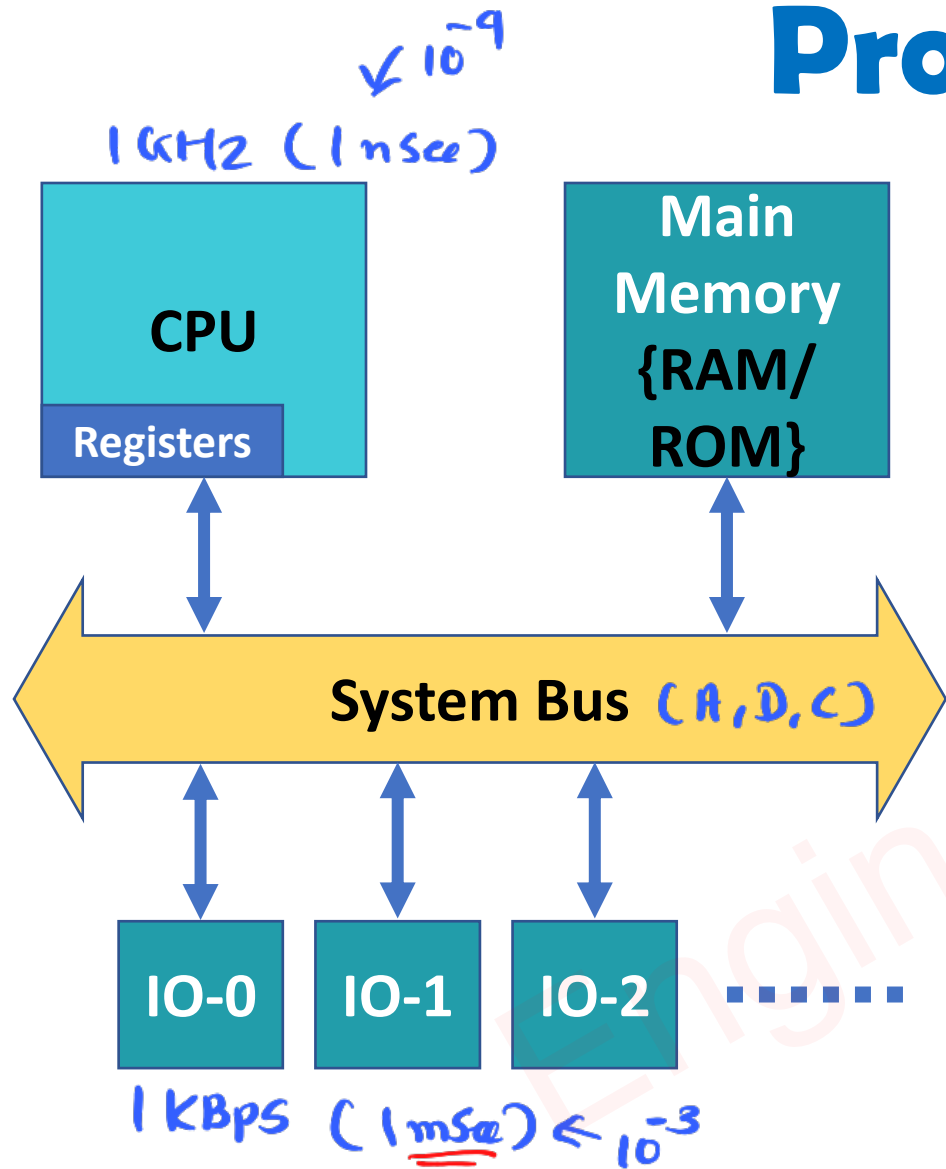


❖ IO Transfer Modes:

1. Programmed IO
2. Interrupt Driven IO
3. DMA

Keyboard, Mouse, Printer, Disk, WIFI, LAN
kHz kHz 10 kHz 5 MHz 100 MHz

Programmed IO



Keyboard, Mouse, Printer, Disk, WIFI, LAN

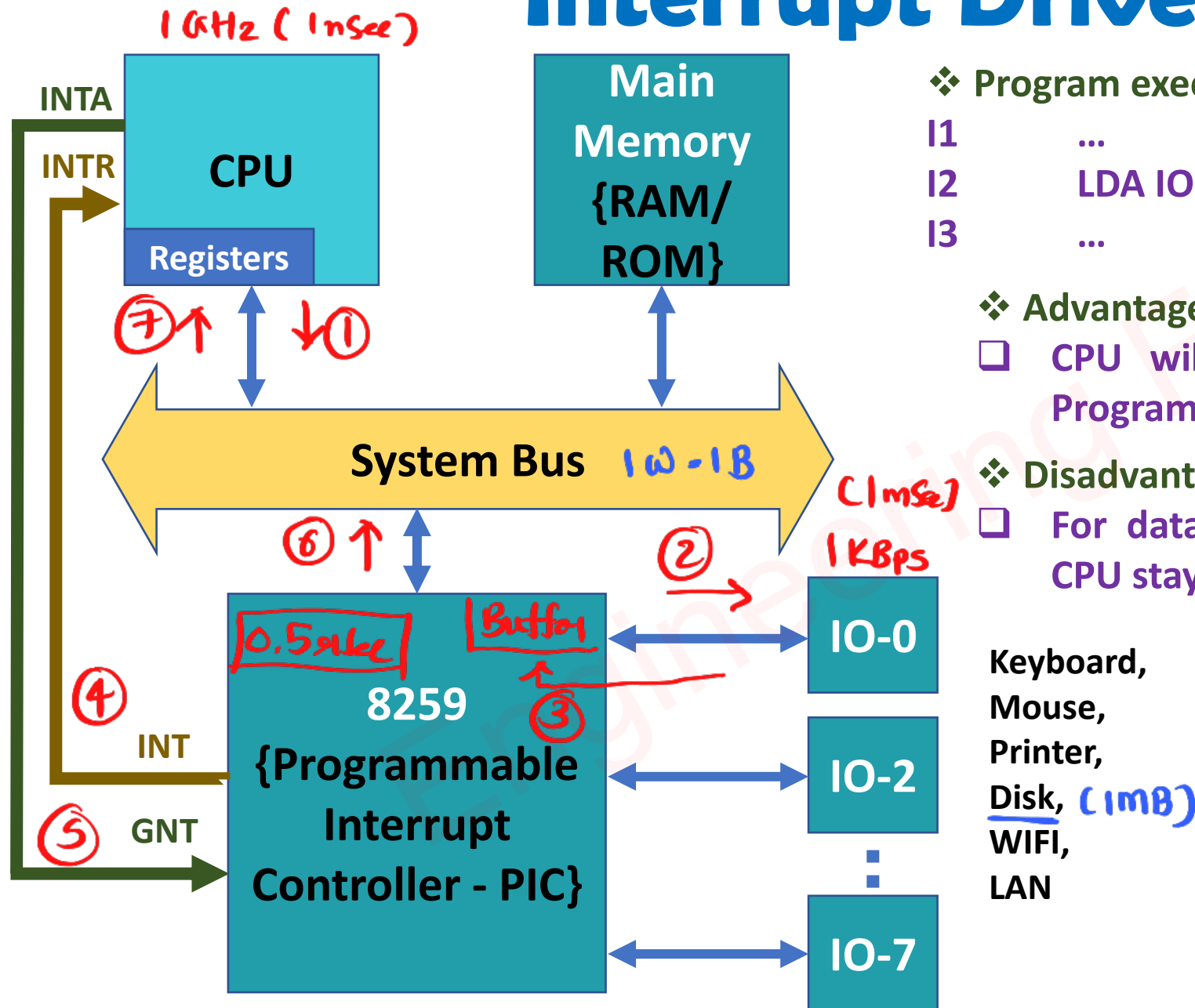
❖ Program execution:

```
I1      ...  
I2      LDA IO-0  
I3      ...
```

❖ Disadvantage:

1. CPU will have more wait time.
2. This type of IO configuration makes performance of CPU slower

Interrupt Driven IO



❖ Program execution:

I1 ...
I2 LDA IO-0
I3 ...

❖ Another Program:

I1 ...
I2 ...
I3 ...

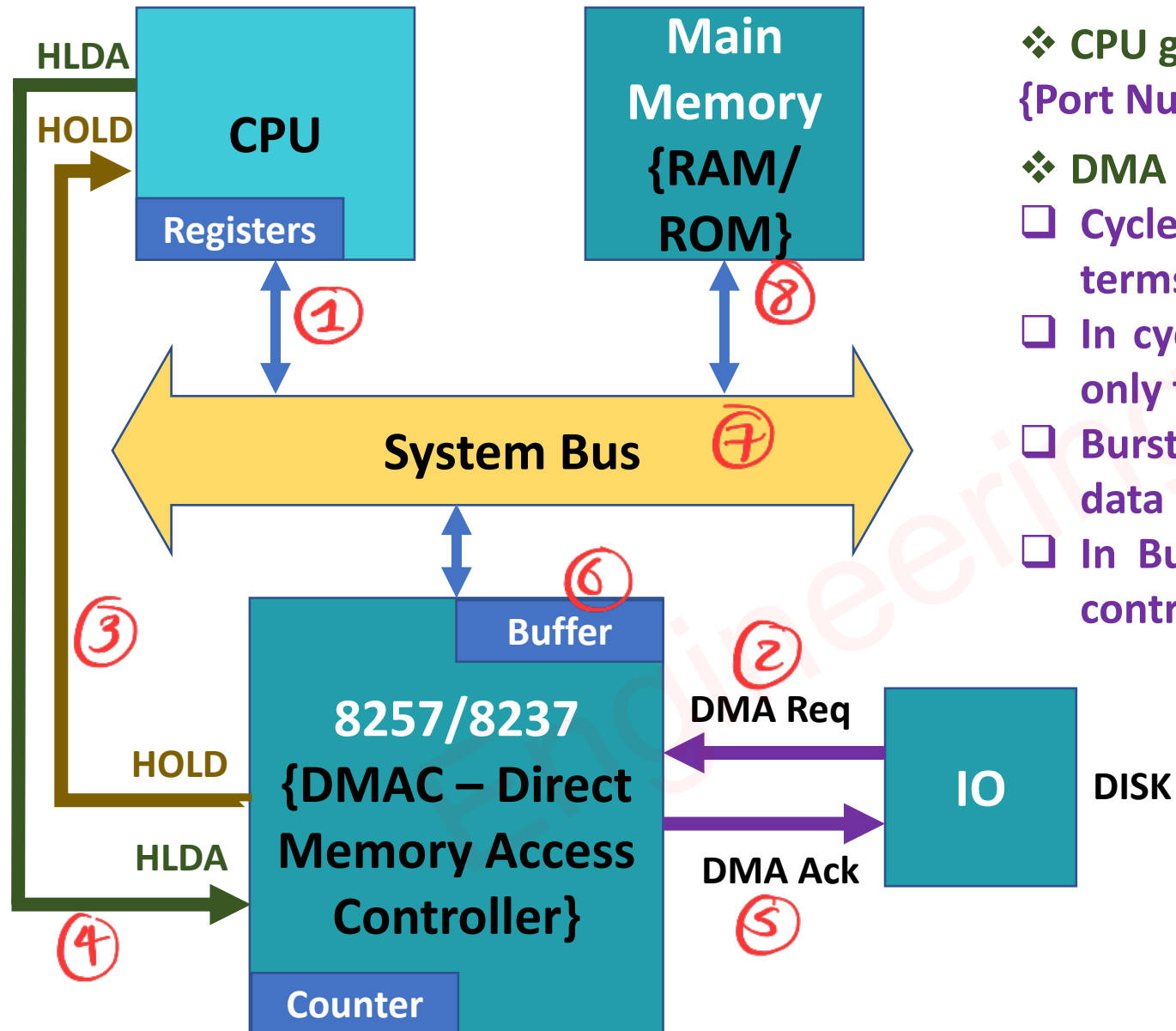
❖ Advantage:

❑ CPU will have less block time compared to Programmed IO.

❖ Disadvantage:

❑ For data transfer in between IO and Memory, CPU stays busy.

DMA – Direct Memory Access



❖ CPU gives IO command at start.

{Port Number, Memory Address, Count, Control Signal}

❖ DMA Modes {It depends on size of buffer}

❑ Cycle stealing: in this mode, DMA sends data in terms of byte by byte or word by word.

❑ In cycle stealing, DMAC waits till buffer is full then only for one cycle only, it takes control of system.

❑ Burst Mode/ Batch Mode: in this mode, DMA sends data in bulk. Here size of buffer is large.

❑ In Burst Mode, once DMAC buffer is full, it takes control of system bus for data transfer.

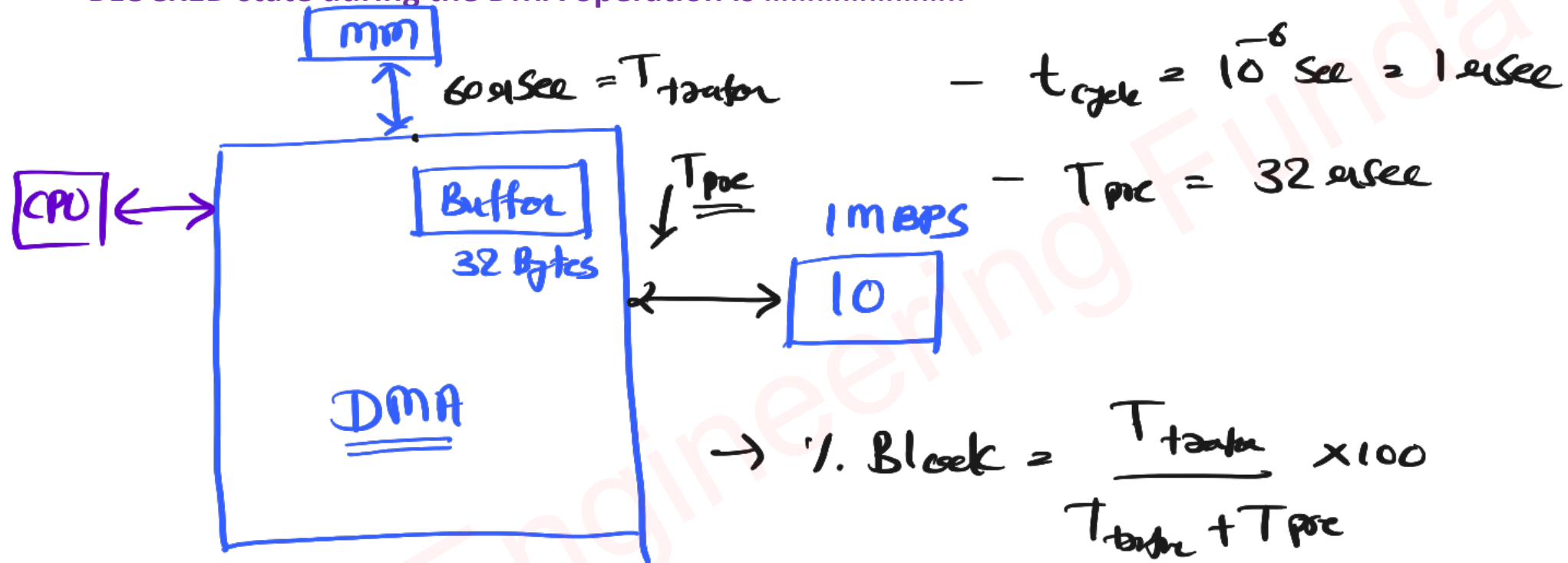
❖ Timings {It depends on size of buffer}

- ❑ Preparation Time: It is time to fill DMA buffer.
- ❑ During preparation time, CPU is free to execute other programs.
- ❑ So during preparation time, CPU is in busy state.
- ❑ Transfer Time: It is time to transfer data from DMA to Main Memory.
- ❑ During Transfer Time, CPU is blocked from system bus.
- ❑ So during transfer time, CPU is in blocked state.

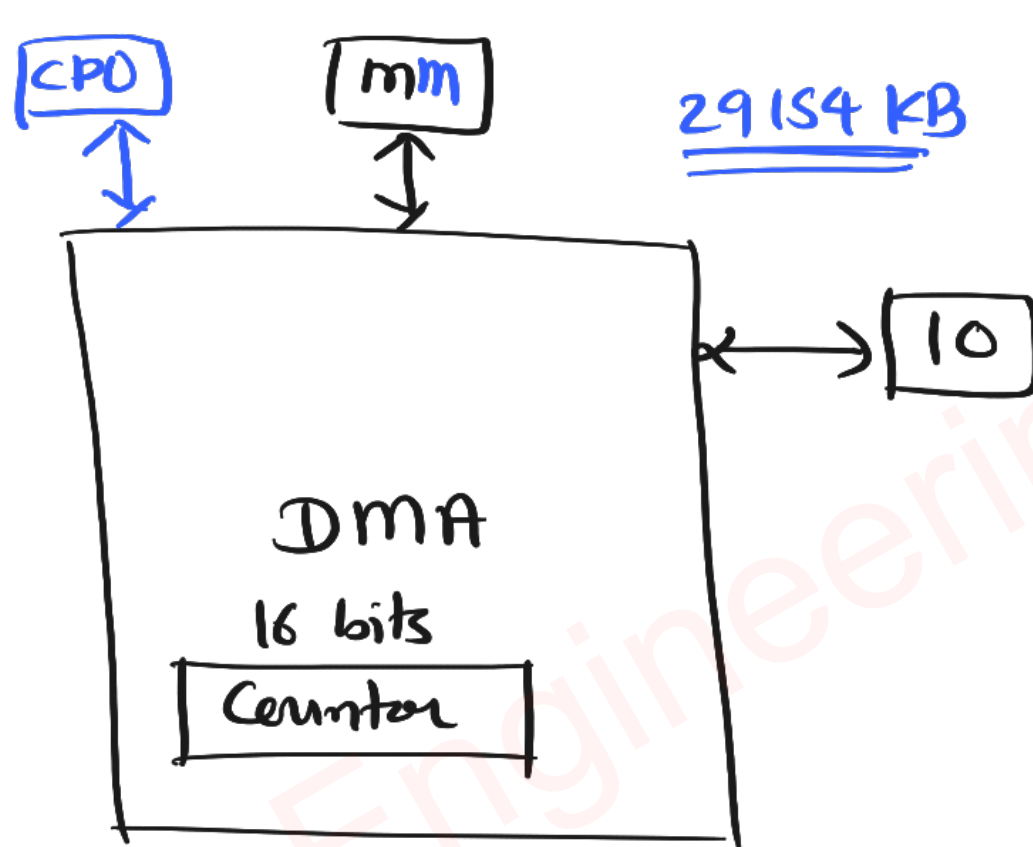
Engineering Funda

Examples on DMA

- ❖ Consider a 1 MBPS IO device interfacing with CPU using DMA in cycle stealing mode. Whenever 32 bytes are present in the buffer, it is transferred to the main memory. Main memory cycle time is $60\mu\text{s}$. Percentage of time CPU is in BLOCKED state during the DMA operation is



- ❖ The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29154 KB from disk to main memory. Here memory is byte addressable. The minimum number of times the DMA controller needs to get control of the system bus from the processor to transfer the file from the disk to main memory is



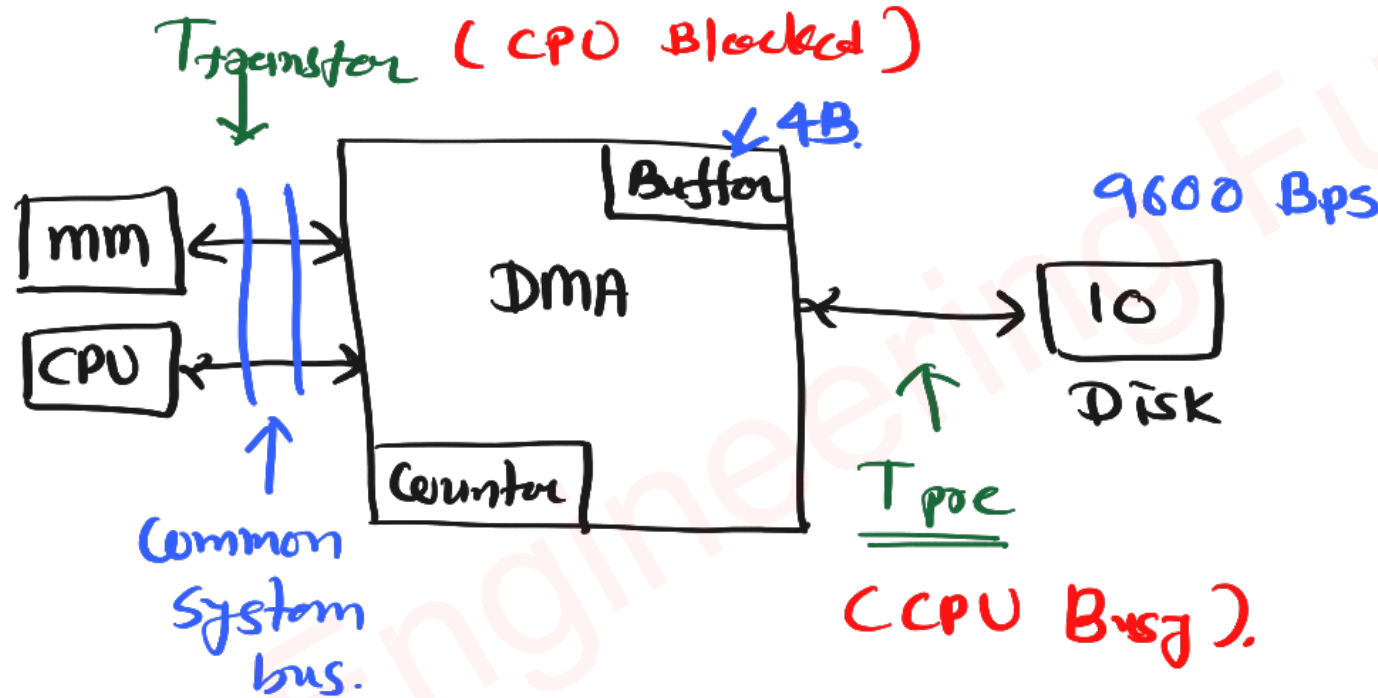
→ In One go data transfer
= 2^{16} B
= 64 KB

→
$$= \frac{29154}{64} = 455.5 = 456$$

Examples on DMA

- ❖ A DMA controller transfers 32 bit words to memory using cycle stealing. The words are assembled from a device that transmits characters at a rate of 9600 characters per second. Assume one character is encoded using 1 byte. The CPU is fetching and executing instructions at an average rate of two million instructions per second. By how much % with the CPU be slowed down because of the DMA transfer?

- A. 0.6%
- ✓ B. 0.12%
- C. 1.2%
- D. 2.5%



$$\Rightarrow T_{pre} = \frac{4B}{9600 \text{ Bps}} = 4.166 \times 10^{-4} \text{ sec}$$

$$\Rightarrow T_{transfer} = \frac{1}{2 \times 10^6} = 0.5 \times 10^{-6} \text{ sec}$$

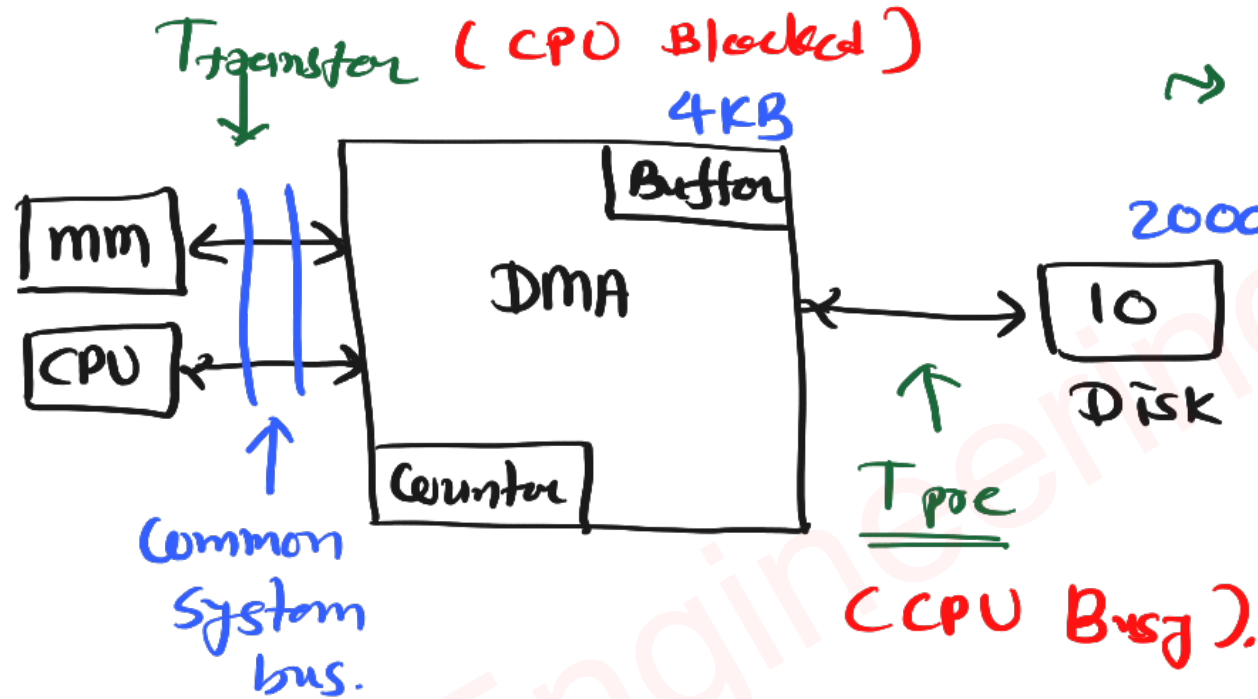
$$\Rightarrow \% \text{ CPU Block} = \frac{T_{transfer}}{T_{transfer} + T_{pre}} \times 100$$

$$= \frac{0.5 \times 10^{-6}}{(4.166 \times 10^{-4} + 0.5 \times 10^{-6})} \times 100 = \underline{\underline{0.12\%}}$$

- ❖ A hard disk is connected to a 50 MHz processor through a DMA controller. Assume that the initial set up of DMA transfer takes 1000 clock cycles for the processor. The hard disk has a transfer rate of 2000 KBPS and average block transferred is 4KB. What fraction of the processor time is consumed by the disk, if the disk is actively transferring 100% of the time.

$$\Rightarrow T_{proc} = \frac{4KB}{2000 KBPS} = 2 \times 10^{-3} \text{ sec}$$

$$\begin{aligned} \Rightarrow T_{transfer} &= 1000 t_{cyc} \\ &= 1000 \times \frac{1}{50 \times 10^6} \\ &= 20 \times 10^{-6} \text{ sec} \end{aligned}$$



$$\begin{aligned} \Rightarrow \% \text{ Block of CPU} &= \frac{T_{transfer}}{T_{transfer} + T_{proc}} \times 100 \\ &= \frac{20 \times 10^{-6}}{20 \times 10^{-6} + 2 \times 10^{-3}} \times 100 = 1\% \end{aligned}$$